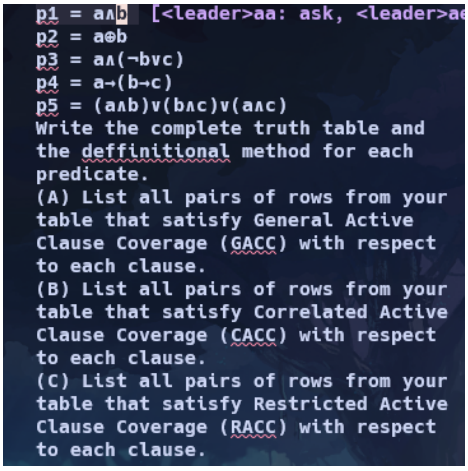
Exercise 10


p1 = a∧b [<leader>aa: ask, <leader>a
p2 = a⊕b
p3 = a∧(¬b∨c)
p4 = a→(b→c)
p5 = (a∧b)∨(b∧c)∨(a∧c)
Write the complete truth table and the deffinitional method for each predicate.
(A) List all pairs of rows from your table that satisfy General Active Clause Coverage (GACC) with respect to each clause.
(B) List all pairs of rows from your table that satisfy Correlated Active Clause Coverage (CACC) with respect to each clause.
(C) List all pairs of rows from your table that satisfy Restricted Active Clause Coverage (RACC) with respect to each clause.

Exercise 10 - Active Clause Coverage

p1 = a ^ b

Truth Table

Row	a	b	p1
1	T	T	T
2	T	F	F
3	F	T	F
4	F	F	F

Definitional Method

Clause a (minor: b):

- p_a=T = T ^ b = b
- p_a=F = F ^ b = F
- a determines p1 when b != F, i.e., **b = T**
- Determining rows: {1, 3}

Clause b (minor: a):

- p_b=T = a ^ T = a
- p_b=F = a ^ F = F
- b determines p1 when a != F, i.e., **a = T**
- Determining rows: {1, 2}

(A) GACC

Major clause	Pairs
a	(1, 3)
b	(1, 2)

(B) CACC

Major clause	Pairs
a	(1, 3)

Major clause	Pairs
b	(1, 2)

(C) RACC

Major clause	Pairs
a	(1, 3)
b	(1, 2)

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p2 = a XOR b

Truth Table

Row	a	b	p2
1	T	T	F
2	T	F	T
3	F	T	T
4	F	F	F

Definitional Method

Clause a (minor: b):

- p_a=T = T XOR b = !b
- p_a=F = F XOR b = b
- a determines p2 when !b != b, which is **always true**
- Determining rows: {1, 2, 3, 4}

Clause b (minor: a):

- p_b=T = a XOR T = !a
- p_b=F = a XOR F = a
- b determines p2 when !a != a, which is **always true**
- Determining rows: {1, 2, 3, 4}

(A) GACC

Major clause	Pairs
a	(1, 3), (1, 4), (2, 3), (2, 4)
b	(1, 2), (1, 4), (2, 3), (3, 4)

(B) CACC

Major clause	Pairs
a	(1, 3), (2, 4)
b	(1, 2), (3, 4)

(C) RACC

Major clause	Pairs
a	(1, 3), (2, 4)
b	(1, 2), (3, 4)

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p3 = a ^ (!b v c)

Truth Table

Row	a	b	c	!b	!b v c	p3
1	T	T	T	F	T	T
2	T	T	F	F	F	F
3	T	F	T	T	T	T
4	T	F	F	T	T	T
5	F	T	T	F	T	F
6	F	T	F	F	F	F
7	F	F	T	T	T	F
8	F	F	F	T	T	F

Definitional Method

Clause a (minors: b, c):

- p_a=T = T ^ (!b v c) = !b v c
- p_a=F = F ^ (!b v c) = F
- a determines p3 when !b v c != F, i.e., **!b v c = T**
- Determining rows: {1, 3, 4, 5, 7, 8}

Clause b (minors: a, c):

- p_b=T = a ^ (F v c) = a ^ c
- p_b=F = a ^ (T v c) = a ^ T = a
- b determines p3 when a ^ c != a, i.e., **a = T and c = F**
- Determining rows: {2, 4}

Clause c (minors: a, b):

- p_c=T = a ^ (!b v T) = a ^ T = a
- p_c=F = a ^ (!b v F) = a ^ !b
- c determines p3 when a != a ^ !b, i.e., **a = T and b = T**
- Determining rows: {1, 2}

(A) GACC

Major clause	Pairs
a	(1, 5), (1, 7), (1, 8), (3, 5), (3, 7), (3, 8), (4, 5), (4, 7), (4, 8)
b	(2, 4)
c	(1, 2)

(B) CACC

Major clause	Pairs
a	(1, 5), (1, 7), (1, 8), (3, 5), (3, 7), (3, 8), (4, 5), (4, 7), (4, 8)
b	(2, 4)
c	(1, 2)

(C) RACC

Major clause	Pairs
a	(1, 5), (3, 7), (4, 8)
b	(2, 4)
c	(1, 2)

---

p4 = a -> (b -> c)

Note: a → (b → c) = !a v (!b v c) = !a v !b v c

Truth Table

Row	a	b	c	b → c	p4 = a → (b → c)
1	T	T	T	T	T
2	T	T	F	F	F
3	T	F	T	T	T
4	T	F	F	T	T
5	F	T	T	T	T
6	F	T	F	F	T
7	F	F	T	T	T
8	F	F	F	T	T

Definitional Method

Clause a (minors: b, c):

- p_a=T = !b v c
- p_a=F = T
- a determines p4 when !b v c != T, i.e., **b = T and c = F**
- Determining rows: {2, 6}

Clause b (minors: a, c):

- p_b=T = !a v c
- p_b=F = !a v T = T
- b determines p4 when !a v c != T, i.e., **a = T and c = F**
- Determining rows: {2, 4}

Clause c (minors: a, b):

- p_c=T = !a v !b v T = T
- p_c=F = !a v !b
- c determines p4 when !a v !b != T, i.e., **a = T and b = T**
- Determining rows: {1, 2}

(A) GACC

Major clause	Pairs
a	(2, 6)
b	(2, 4)
c	(1, 2)

(B) CACC

Major clause	Pairs
a	(2, 6)
b	(2, 4)
c	(1, 2)

(C) RACC

Major clause	Pairs
a	(2, 6)
b	(2, 4)

Major clause	Pairs
c	(1, 2)

p5 = (a ^ b) v (b ^ c) v (a ^ c)

Truth Table

Row	a	b	c	a ^ b	b ^ c	a ^ c	p5
1	T	T	T	T	T	T	T
2	T	T	F	T	F	F	T
3	T	F	T	F	F	T	T
4	T	F	F	F	F	F	F
5	F	T	T	F	T	F	T
6	F	T	F	F	F	F	F
7	F	F	T	F	F	F	F
8	F	F	F	F	F	F	F

Definitional Method

Clause a (minors: b, c):

- p_a=T = (T ^ b) v (b ^ c) v (T ^ c) = b v (b ^ c) v c = b v c
- p_a=F = (F ^ b) v (b ^ c) v (F ^ c) = b ^ c
- a determines p5 when (b v c) != (b ^ c), i.e., **b != c**
- Determining rows: {2, 3, 6, 7}

Clause b (minors: a, c):

- p_b=T = a v c v (a ^ c) = a v c
- p_b=F = (a ^ c)
- b determines p5 when (a v c) != (a ^ c), i.e., **a != c**
- Determining rows: {2, 4, 5, 7}

Clause c (minors: a, b):

- p_c=T = (a ^ b) v b v a = a v b
- p_c=F = (a ^ b)
- c determines p5 when (a v b) != (a ^ b), i.e., **a != b**
- Determining rows: {3, 4, 5, 6}

(A) GACC

Major clause	Pairs
a	(2, 6), (2, 7), (3, 6), (3, 7)
b	(2, 4), (2, 7), (5, 4), (5, 7)
c	(3, 4), (3, 6), (5, 4), (5, 6)

(B) CACC

Major clause	Pairs
a	(2, 6), (2, 7), (3, 6), (3, 7)
b	(2, 4), (2, 7), (5, 4), (5, 7)
c	(3, 4), (3, 6), (5, 4), (5, 6)

(C) RACC

Major clause	Pairs
a	(2, 6), (3, 7)
b	(2, 4), (5, 7)
c	(3, 4), (5, 6)